

S2Q27

Question 27 – Stage 2

A suspended sphere of ice is melting in such a way that its radius is decreasing constantly by 1mm per day, whilst retaining its spherical shape.

The water drips into a container that is in the form of a regular tetrahedron with a vertex on the ground and its upper open face parallel to the ground.

The sphere initially has a radius of 1 metre and the tetrahedral container is empty.

At the moment when the radius is 99cm what is the rate of change (in mm/hour to 4.sig. figs) of the height of water in the container?

Working

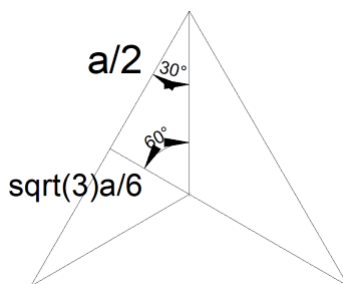
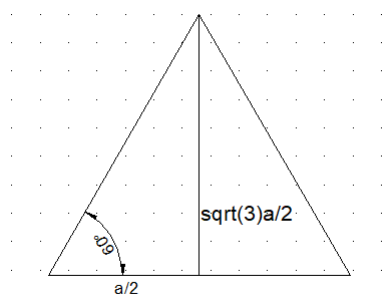
$$\frac{dr}{dt} = -\frac{1}{24} \text{ mm/hour}$$

$$\frac{dV_s}{dr} = 4\pi r^2$$

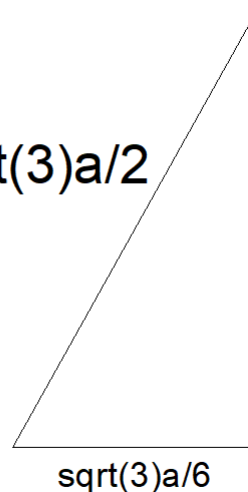
$$\frac{dV_s}{dt} = -\frac{1}{24} \cdot 4\pi r^2$$

$$= -\frac{\pi r^2}{6}$$

$$\frac{dV_t}{dt} = \frac{\pi r^2}{6}$$



$$\sqrt{3}a/2$$



$$V_t = a^3 \frac{\sqrt{2}}{12}$$

$$h = \sqrt{\left(\frac{\sqrt{3}a}{2}\right)^2 - \left(\frac{\sqrt{3}a}{6}\right)^2}$$

$$h = \frac{a\sqrt{6}}{3}$$

$$\frac{\sqrt{6}}{2}h = a$$

$$V_t = \left(\frac{\sqrt{6}}{2}h\right)^3 \frac{\sqrt{2}}{12}$$

$$= \frac{\sqrt{2}}{12} \cdot \frac{3\sqrt{6}}{4} h^3$$

$$= \frac{\sqrt{3}}{8} h^3$$

$$\frac{8\sqrt{3}}{3} V_t = h^3$$

$$h = \left(\frac{8\sqrt{3}}{3} V_t\right)^{\frac{1}{3}}$$

$$= \left(\frac{8\sqrt{3}}{3}\right)^{\frac{1}{3}} V_t^{\frac{1}{3}}$$

$$\frac{dh}{dV_t} = \frac{1}{3} \left(\frac{8\sqrt{3}}{3}\right)^{\frac{1}{3}} V_t^{-\frac{2}{3}}$$

$$\frac{dh}{dt} = \frac{1}{3} \left(\frac{8\sqrt{3}}{3} \right)^{\frac{1}{3}} V_t^{-\frac{2}{3}} \cdot \frac{\pi r^2}{6}$$

$$V_t = \frac{4}{3} \pi (1000^3 - r^3)$$

$$\frac{dh}{dt} = \frac{1}{3} \left(\frac{8\sqrt{3}}{3} \right)^{\frac{1}{3}} \left(\frac{4}{3} \pi (1000^3 - r^3) \right)^{-\frac{2}{3}} \cdot \frac{\pi r^2}{6}$$

≈ 1.143